

plant to the early stages of commercial production. All of this costs money in varying amounts. All of it is a drain on other profits of the business. Even in the early stage of commercial production the extra sales expense involved in building sufficient volume for a new product to furnish the desired margin of profit is such that the out-of-pocket losses at this stage of development may be greater than they were during the pilot-plant period.

From the standpoint of the investor there are two aspects of all this that have particular significance. One of these is the impossibility of depending on any sure time table in the development cycle of a new product. The other is that even for the most brilliantly-managed enterprises, a percentage of failures is part of the cost of doing business. In a sport, such as baseball, even the most outstanding league champions will have dropped some percentage of their scheduled games.

The point in the development of a new process that is perhaps worth the closest scrutiny from the standpoint of timing the buying of common stocks is that at which the first full-scale commercial plant is about to begin production. In a new plant for even established processes or products, there will probably be a shake-down period of six to eight weeks that will prove rather expensive. It takes this long to get the equipment adjusted to the required operating efficiency and to weed out the inevitable "bugs" that seem to occur in breaking in modern intricate machinery. When the process is really revolutionary, this expensive shake-down period may extend far beyond the estimate of even the most pessimistic company engineer.

Furthermore, when problems finally do get solved, the weary stockholder still cannot look forward to immediate profits. There are more months of still further drain while even more of the company's profits from older lines are being ploughed back into special sales and advertising efforts to get the new product accepted.

It may be that the company making all this effort is having such growth in revenue from other and older products that the drain on profits is not noticed by the average stockholder. Frequently, however, just the opposite happens. As word first gets out about a spectacular new product in the laboratory of a well-run company, eager buyers bid up the price of that company's shares. When word comes of successful pilot-plant operation, the shares go still higher. Few think of the old analogy that operating a pilot plant is like driving an automobile over a winding country road at ten miles